

CinemaPulse

Audience Segmentation for Personalized OTT Content Recommendation

1. Problem Statement

OTT platforms serve millions of users with different viewing preferences. Recommending the same content to every user reduces engagement and user satisfaction.

The objective of this project is to automatically group users with similar viewing behavior into audience segments using Machine Learning. These segments help OTT platforms provide personalized content recommendations and improve user engagement.

Why is this important?

- Improves personalized recommendations
- Increases user retention
- Supports targeted marketing campaigns
- Helps understand audience behavior
- Enhances overall user experience

2. Proposed Solution

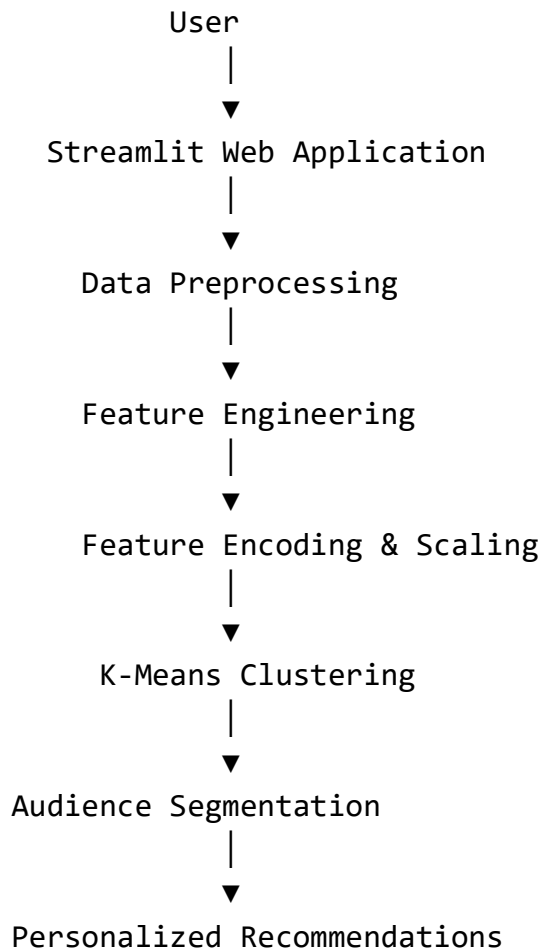
This project uses **K-Means Clustering**, an unsupervised Machine Learning algorithm, to segment OTT users based on their behavior and preferences.

The application:

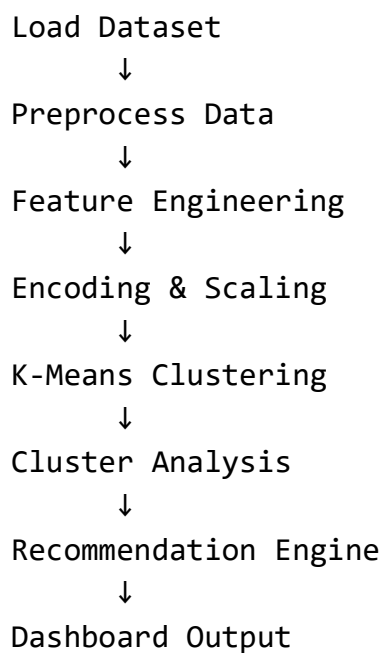
- Loads the user dataset
- Cleans and preprocesses the data
- Performs feature engineering
- Groups similar users into audience segments
- Generates personalized content recommendations
- Displays results through an interactive Streamlit dashboard

3. System Architecture

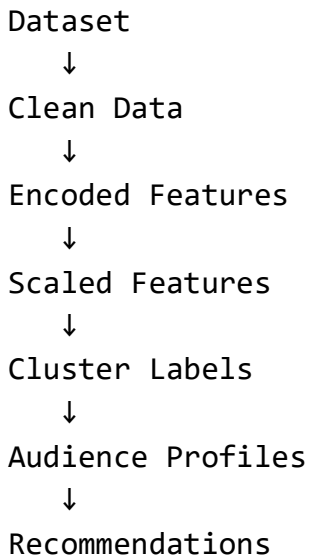
Architecture Diagram



Workflow



Data Flow



4. Technology Stack

Programming Language:	Python
Frontend:	Streamlit
Machine Learning:	Scikit-learn
Data Processing:	Pandas, NumPy
Visualization:	Plotly, Matplotlib
Model Saving:	Joblib
Version Control:	Git & GitHub

5. Machine Learning Workflow

Dataset

The project uses an OTT User Dataset [netflix_users.csv] containing:

- Age
- Country
- Subscription Type
- Watch Time
- Favorite Genre
- Last Login

Data Preprocessing

- Handle missing values
- Remove duplicates
- Convert Last_Login into Days_Since_Last_Login
- Remove unnecessary columns (User_ID, Name)

Feature Engineering

- Create Days_Since_Last_Login
- Encode categorical features
- Scale numerical features using StandardScaler

Model Training

- Determine optimal clusters using Elbow Method and Silhouette Score
- Train the K-Means model
- Assign each user to a cluster

Prediction

The trained model predicts the audience segment for each user and generates personalized recommendations.

6. Algorithm Selection

Selected Algorithm

K-Means Clustering

Why K-Means?

- Suitable for unlabeled data
- Fast and efficient
- Easy to interpret
- Creates meaningful audience segments
- Widely used for customer segmentation

7. Evaluation Metrics

Since this is an **Unsupervised Learning** problem, classification metrics like Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC are **not applicable**.

The model is evaluated using:

- Elbow Method
- Silhouette Score

8. Future Enhancements

- Real-time recommendation system
- Hybrid recommendation engine
- Cloud deployment
- User authentication
- Deep Learning-based segmentation
- Collaborative filtering

9. Conclusion

CinemaPulse successfully segments OTT users based on their viewing behavior and preferences using Machine Learning. The generated audience segments enable personalized recommendations, helping OTT platforms improve engagement, user retention, and content delivery.